

Software Testing & Security Checking

EE 5191

Syllabus

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<http://cc.ee.ntu.edu.tw/ST>

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Teaching Plan

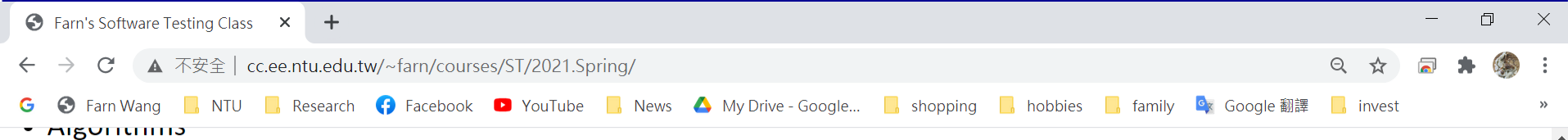
With the availability of lecture videos, class lecture of each week is administered as follows.

- Presenting the guideline to the materials
- showing some videos
- Q&A sessions
- a quiz (open book and open internet)

We expect the students to prepare well enough for the quizzes with the slides and videos online.

Teaching Plan

Video links in the course webpage.



Videos of lectures for each week:

- Videos for the 1st class (2021/02/24):

[Syllabus](#), [module 1.1](#),

- Videos for the 2nd class (2021/03/03):

[module 1.2](#), [module 1.3](#), [module 1.4](#),

- Videos for the 3rd class (2021/03/10):

[Module 1.5](#), [Module 1.6 & Assignment of market survey](#), [Module 1.7](#),

- Videos for the 4th class (2021/03/17):

[Module 2.1](#), [Module 2.2](#), [Module 2.3](#)

- Videos for the 5th class (2021/03/24):

授課資訊

- 課號：EE 5191
 - Spring 2020
 - Wednesday, 9:10am-12:00noon
- Classroom: BL 113 Credits: 3
- 王凡 教授
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 - farn@ntu.edu.tw
 - <http://cc.ee.ntu.edu.tw/~farn/courses/ST>
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Background

- Software projects have spent more than half of their budget in testing and debugging.
- Software security is a highly visible and fast-growing sector of software testing.
 - government regulations
 - companies willing to pay
- Taiwan's high-tech companies do not think that they should spend money in software testing.
 - Taiwan's industrial strength in OEM and ODM
- Without the know-how in software QA, Taiwan has no hope.

Types of Software Testing

- **Functional Testing**

- **Unit testing**
 - **Integration testing**
 - **End-to-end testing**
 - **Smoke testing**
 - **Sanity testing**
 - **Regression testing**
 - **Acceptance testing**
 - **Interface testing**
- **White box testing**
 - **Black box testing**

Types of Software Testing

- **Non-functional Testing**

- Performance testing
- Security testing
- Load testing
- Failover testing
- Compatibility testing
- Usability testing
- Scalability testing
- Volume testing
- Stress testing
- Accessibility testing
- Alpha testing
- Beta testing
- Maintainability testing
- Compliance testing
- Efficiency testing
- Reliability testing
- Endurance testing
- Disaster recovery testing
- Localization testing
- Internationalization testing
- Ad-hoc testing

產業分析

Testing costs

- 20 percent of the overall development time for a single-component application,
- 20 to 30 percent for a two-component application and
- 30 to 35 percent for an application with GUI.
- 35 to 50 percent for a distributed application with GUI.



產業分析

Global Test automation Market Revenue, 2015 - 2021
(USD Billion)



Source: Zion Market Research 2017

軟體自動測試市場主要競爭者說明

- Open-Source Solutions

<https://phoenixnap.com/blog/best-automation-testing-tools>

Tools	Com	web	And.	iOS	GUI	API	src	AI	TRG shop
TaaD		√	√	√	√	X	X	√	√
Citrus	Telerik	√	X	X	X	√	X	X	X
Galen		√	X	X	X	X	X	X	X
Karate-DSL		√	X	X	X	√	X	X	X
Robot Framework		√	√	√	X	√	√	X	X
Selenium		√	X	X	√	√	√	X	X
Watir	MS	√	X	X	X	X	√	X	X
Gauge (β)	HP	√	√	√	X	√	√	X	X
RedwoodHQ		√	X	X	√	√	X	X	X
Serenity BDD		√	X	X	√	X	X	X	X
Appium		X	√	√	√	X	X	X	X
Apache JMeter		√	X	X	X	√	X	X	X

軟體自動測試市場主要競爭者

- vs. Commercial Solutions

<https://phoenixnap.com/blog/best-automation-testing-tools>

Tools	Com	\$\$	web	And.	iOS	GUI	API	src	AI	TRG Shop
TaaD		6000/y	√	√	√	√	X	X	√	√
Rat'al Funct'al Tester	IBM	8140/y	√	√	√	√	√	√	X	X
Katalon Studio		759/y	√	√	√	√	√	√	X	X
Ranorex		1120/y	√	√	√	√	√	√	√	X
Sahi Pro		1390/y	√	X	X	√	X	X	X	X
TestComplete		4515/y	√	√	√	√	√	√	√	X
TestPlant eggPlant	MS		√	√	√	√	√	√	X	X
Tricentis Tosca	HP		√	√	√	√	√	√	X	X
Unif. Funct'al Testing		3200/y	√	X	X	X	√	X	√	X
Cypress		1188/y	√	X	X	X	√	X	X	X

軟體自動測試市場主要競爭者年營收

年營收
Revenue

M : 1 百萬美元

- QualityLogic, <https://www.qualitylogic.com/> (9.7M)
- V-Cubed Solutions, <https://www.v-cubed.net/> (27M)
- Value Stream Engineering, <http://www.valuestreamengineers.com/> (3.8M)
- IT Automation LLC, <http://www.itautomation.com/> (8.8M)
- NeoTys, <https://www.neotys.com/> (8.8M)
- Parasoft, <https://www.parasoft.com/> (55M)
- Performancelab, <https://www.performancelab.com.br/home> (25.3M)
- ScienceSoft, <https://www.scnsoft.com/> (22M)
- Symbio, <https://www.symbio.com/> (150M)
- tenio.io, <https://tenon.io/>

Avg. 34.5 Millions
~310.4 Millions

軟體自動測試市場主要競爭者說明

- 您看見機會了嗎？

Huge opportunity!

Global Test Service Market 2015 - 2021

310.71 (USD billion)

of testing service market captured by top 9.



Source: Zion Market Research 2017

產業分析

from the Boss' perspective

Once upon a time in a QA meeting when MS hired 10000 testers,

“I know I pay a lot of testers, but I really have no idea what they do.”



<https://softwaretesting.cioreview.com/cxoinsight/the-cost-of-software-testing-nid-27746-cid-112.html>

產業分析

from Taiwan Boss' perspective

If our product does not
crash before clients, all
your testing is wasted.



產業分析

from the QA engineers' perspective

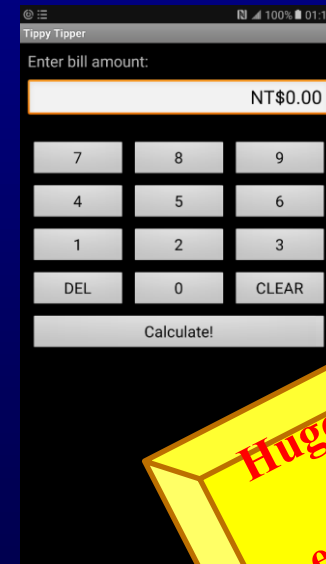
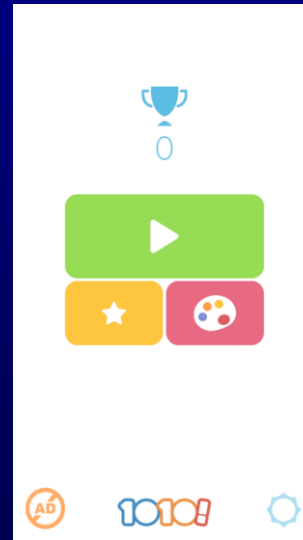
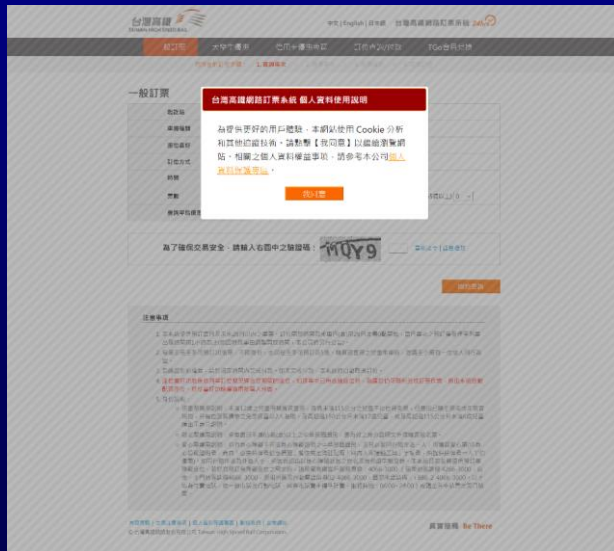
I am a loser.
I don't have value to the
company.
I have no life!



- Can work in a big company!
- Can test a few products in the company.
- Low self-esteem.

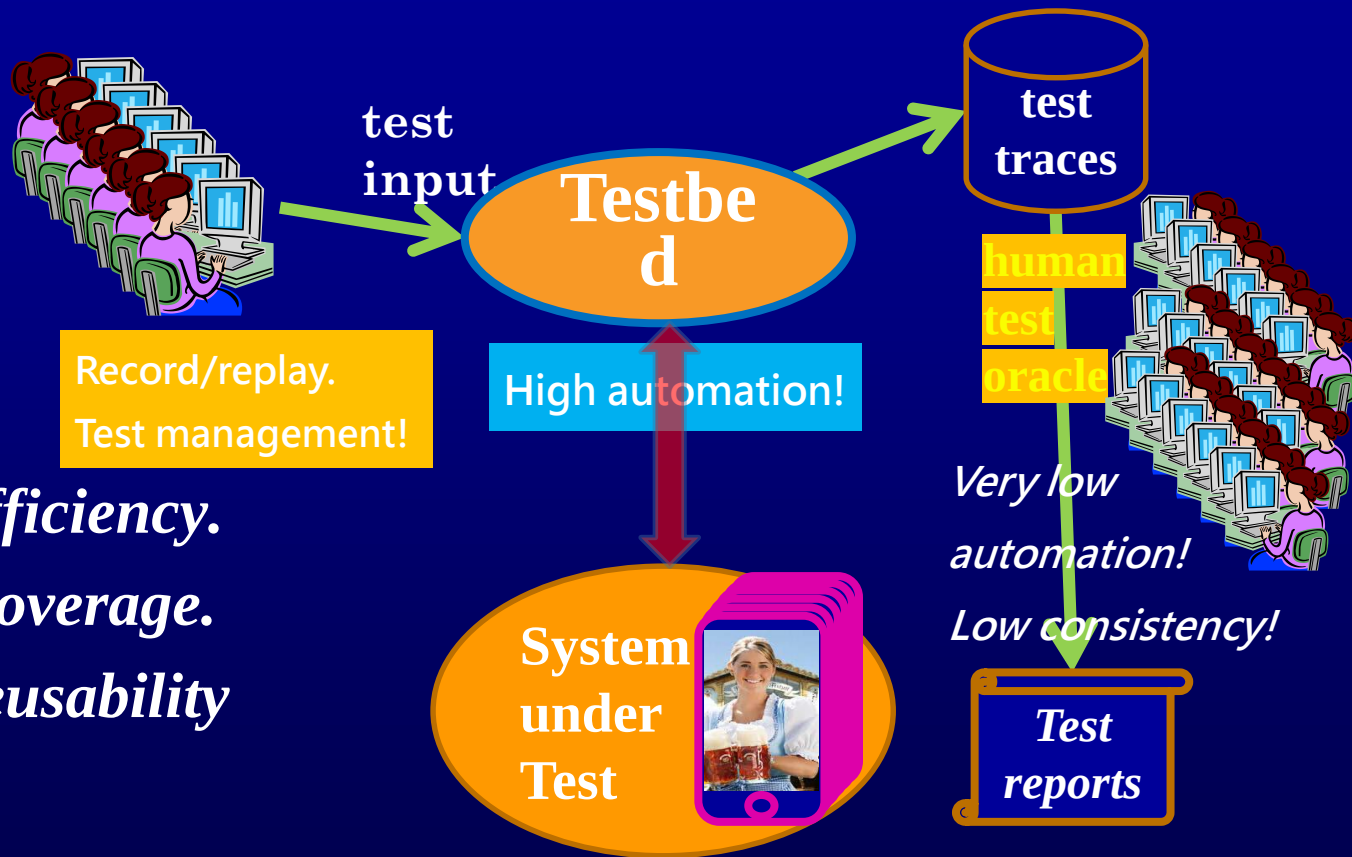
產業分析 Reality

- *Basically record/replay and test management.*
- *Need human to design tests*
- *Need human to recognize screens*
- *Need human to understand traces.*



**Huge labor
low
efficiency
inconsistent
quality**

產業分析 Reality



- *Low efficiency.*
- *Poor coverage.*
- *Low reusability*

產業分析

solutions nowadays

- 測試外包
 - 適合大型軟體公司
 - 只能做system test或user acceptance testing
 - IP外洩風險
 - Feedback 緩慢
- 軟體測試自動化工具
 - 只幫你record與replay
 - Monkey testing (low intelligence)
 - Test investment low reusability
 - 許多未經整合的測試準備工作



★★巨大市場機會★★
App智慧型測試工具

產業分析

- 軟體品管預算膨脹迅速
 - 目前一般軟體開發計畫，50%以上成本與時間都花在測試與品管。
 - 若能降低10%軟體品管成本，就可提昇公司5%獲利。
- 軟體品管技術交流不易
 - 空有天下優良軟體工程師，卻只能進行手工測試。
 - 目前一般透過網路廣告，測試需求顧客與軟體測試服務公司個別接洽！
 - 底層結構（**infrastructure**）把持在少數測試工具公司！

產業分析

- 應用測試難以自動化
 - 目前測試自動化成果重複使用性都非常低
 - 對一個顧客所製作的成果，幾乎都沒辦法用在另一個顧客。
 - 軟體流程差異性、軟體文件差異性
 - 學界的軟體測試成果基本上沒有在工業界應用可能性。

Goal

- Theory in software testing.
- Advanced testing and security-checking tools
- Capability of deploying software testing technology to various companies.
- Basic concepts and skills in security checking.

Course Plan

- The first module: foundations
- The second module: Testing criteria
 - Graph testing
 - Logic testing
 - Input space testing
 - Syntax testing
- Third module: practical consideration.
- Criterias for new app types
- Security checking overview.

Pre-requisites

- Basic Computer Concepts
- C/C++, Java programming
- Python
- Data-Structures
- Algorithms

Evaluation

- Final exam (40%) , June 23, 2021
- Mid-term report (8%), deadline on April 28, 2021.
- A Software Testing Market Survey Report (2%)
 - Deadline March 17, 2021
- Quizzes (10%), open book, open internet, no discussion
- 4 projects (40%)

Evaluation

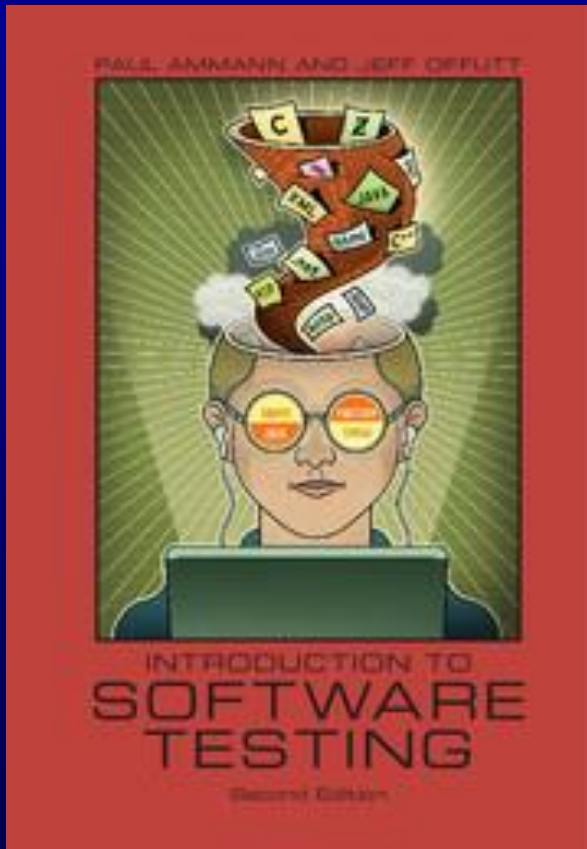
- Midterm presentation (8%)
 - The topic must be technologically related to one of the chapter in the textbook.
 - a main paper
 - ◆ must be a regular paper published in ACM/IEEE ICSE 2019-20, ACM FSE 2019-20, IEEE ASE 2019-20, ACM ISSTA 2019-20, or ACM PLDI 2019-20.
 - 2 comparison papers
 - Comparing the three papers that you selected in the above.

Evaluation

- 4 projects (40%):
 - Project 1 (10%): Continuous testing and unit testing,
 - due by April 7, 2021
 - Project 2 (10%): GUI testing with Selenium
 - with Firefox on either Linux or Windows
 - due by May 5, 2021
 - Project 3 (10%): mobile app testing
 - due by May 26, 2021
 - Project 4 (10%): security check
 - due by June 16, 2021

Textbook

Introduction to Software Testing



Paul Ammann and Jeff Offutt

related materials in

<http://cs.gmu.edu/~offutt/softwaretest/>

Course Schedule (1/2)

1.	2/24	a) Syllabus, b) Chapter 1
2.	3/3	Chapter 1
3.	3/10	a) Chapter 1 b) Survey assignment announcement c) Deadline for midterm presentation registration
4.	3/17	a) Chapter 2 b) Survey assignment deadline c) Confirmation of midterm presentation registration d) Project 1 announcement: <i>unit testing</i>
5.	3/24	Chapter 2
6.	3/31	Chapter 2
7.	4/7	a) Chapter 2b) Project 1 deadline c) Project 2 announcement: <i>web-testing</i>
8.	4/14	Chapter 3
9.	4/21	Chapter 3

Course Schedule (2/2)

10.	4/28	a) Deadline of midterm presentation submission b) Midterm presentation 1
11.	5/5	a) Chapter 3 b) Project 2 deadline c) Project 3 announcement: mobile app testing
12.	5/12	Midterm presentation 2
13.	5/19	Chapter 4
14.	5/26	a) Chapter 5b) Project 3 deadline c) Project 4 announcement: security checking
15.	6/2	Chapter 5
16.	6/9	Security checking
17.	6/16	a) Security checking b) Project 4 deadline
18.	6/23	Final exam